

Uniting theorems:

1. $XY + XY' = X$

Absorption theorems:

2. $X + XY = X$

Elimination theorems:

3. $X + X'Y = X + Y$

Duality:

4. $(X + Y + Z + \dots)' = XYZ \dots$

Theorems for multiplying out and factoring:

5. $(X + Y)(X' + Z) = XZ + X'Y$

Consensus theorems:

6. $XY + YZ + X'Z = XY + X'Z$

Operations with 0 and 1:

1. $X + 0 = X$

2. $X + 1 = 1$

Idempotent laws:

3. $X + X = X$

Involution law:

4. $(X')' = X$

Laws of complementarity:

5. $X + X' = 1$

Commutative laws:

6. $X + Y = Y + X$

Associative laws:

7. $(X + Y) + Z = X + (Y + Z)$
 $= X + Y + Z$

Distributive laws:

8. $X(Y + Z) = XY + XZ$

DeMorgan's laws:

9. $(X + Y)' = X'Y'$

Exclusive OR

$X \oplus Y = X'Y + XY'$

1D. $(X + Y)(X + Y') = X$

2D. $X(X + Y) = X$

3D. $X(X' + Y) = XY$

4D. $(XYZ \dots)' = X + Y + Z + \dots$

5D. $XY + X'Z = (X + Z)(X' + Y)$

6D. $(X + Y)(Y + Z)(X' + Z) = (X + Y)(X' + Z)$

1D. $X \cdot 1 = X$

2D. $X \cdot 0 = 0$

3D. $X \cdot X = X$

5D. $X \cdot X' = 0$

6D. $XY = YX$

7D. $(XY)Z = X(YZ) = XYZ$

8D. $X + YZ = (X + Y)(X + Z)$

9D. $(XY)' = X' + Y'$

Equivalence

$(X \equiv Y) = XY + X'Y'$

(L) closed $V_{in} = 0$ (L)
can sink current without generating voltage

(H) open
 $V_{in} = V_{cc} - I_{IH}R$
 $V_{in} > V_{IHmin}$

$$V_{cc} - I_{IH}R > V_{IHmin}$$

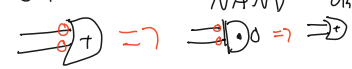
$$\frac{V_{cc} - V_{IHmin}}{I_{IHmax}} > R$$

(L) closed $V_{in} = V_{cc}$
can source current w/o voltage drop

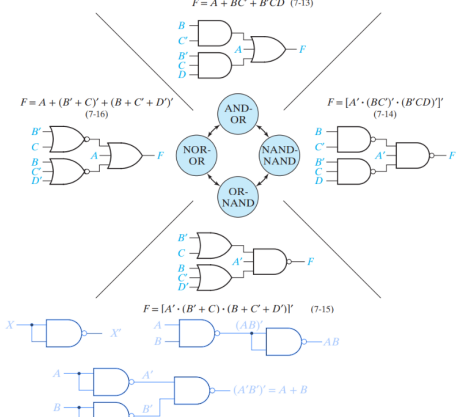
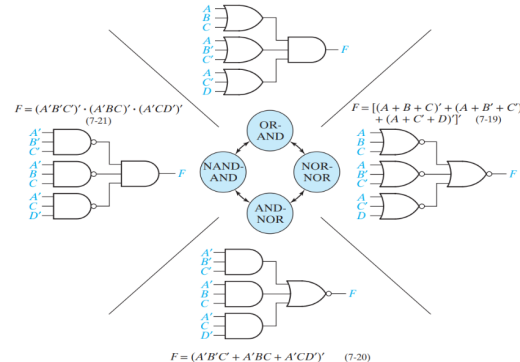
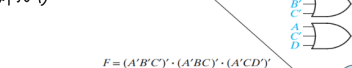
(H) open
 $V_{in} = R I_{IL}$
 $V_{in} < V_{ILmax}$
 $R I_{IL} < V_{ILmax}$

$$R < \frac{V_{ILmax}}{I_{ILmax}}$$

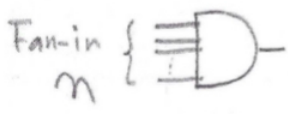
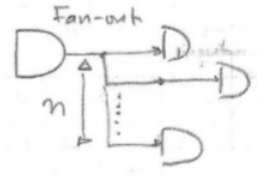
Complement input to OR



Complement inputs to AND

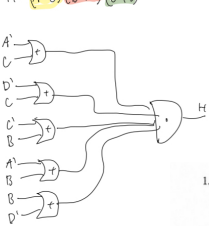
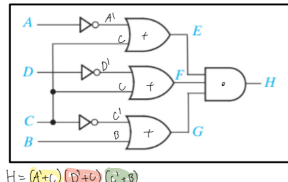
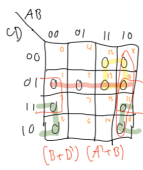
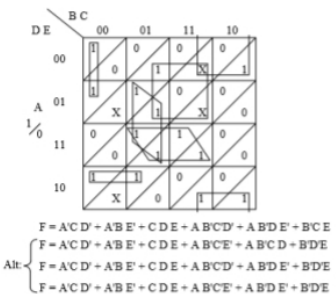
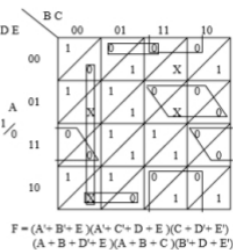
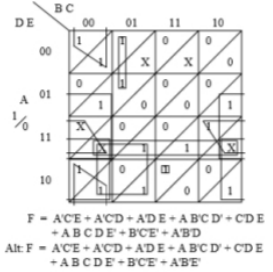
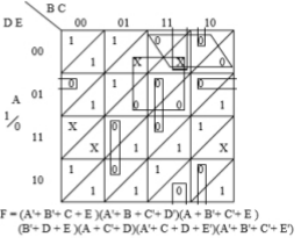


$x_2 \rightarrow x_{16}$: group in 4's
 $x_2 \rightarrow x_8$: group in 2's
 $x_4 \rightarrow x_8$: Keep doubles by 10
in 2's comp, subtract the place
ex 110001 $\rightarrow$ 1's
1 1 0 0 0 1
 $\rightarrow$ 1 1 0 0 0 1



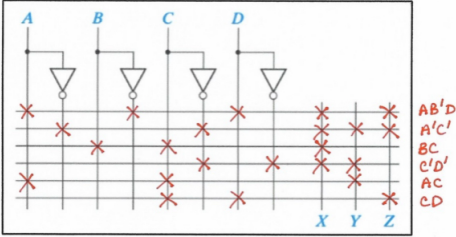
Fan-out $L = n$
Fan-in $H = m$
Fan-out $= \min(m, n)$

$m < \frac{I_{OHmax}}{I_{IHmax}}$
 $n < \frac{I_{OLmax}}{I_{ILmax}}$

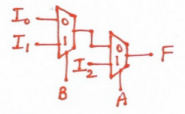


3. Make the necessary connections in the PLA below to implement the following equations:

$X = AB'D + A'C' + BC + C'D'$
 $Y = A'C' + A'C + C'D'$
 $Z = C'D + A'C' + AB'D$



1. Show how two 2-to-1 multiplexers (with no added gates) could be connected to form a 3-to-1 MUX. Input selection should be as follows:
If $AB = 00$, select I_0
If $AB = 01$, select I_1
If $A = 1$, (B is a don't-care), select I_2 .



2. Show how to make a 4-to-1 MUX using four three-state buffers and a decoder.

